Jaine Turban Kinetic Energy Tile 5/7/25

Introduction:

The Kinetic Energy Tile(KET) is a project demonstrating how the basic principles of Physics mechanics can lead to a sustainable energy source. As demand for electricity increases, the need for coal and fossil fuels increases. However, the former is becoming increasingly limited and has ethical consequences. A better way to harvest energy is passively, such as generating electricity as millions of New Yorkers commute using kinetic energy tiles.

Project Design Process:

Electrical Generators are devices that convert mechanical energy to electrical energy. One common generator is a DC motor. As the shaft in a DC motor is rotated, electricity is induced due to Copper coils. Thus, to produce electrical energy from footsteps, the linear motion resulting from footsteps must be converted to rotational motion on the shaft of a DC motor. To achieve this, 4 main subsystems were created. These 3 subsystems are: “Base”, “Rack & Pinion”, “Pulley system”, and “Output gear”.

Base:

The base(as depicted below/above) needed to be designed to withstand the force exerted by a person’s footstep and restore to its original state prior to the step. Consequently, the base was designed with 4 springs to evenly distribute weight. The restorative force of the spring, $F_s = -kx$, allows the base to return to its original state proportional to the weight applied. The springs used within the design have a spring constant(k) of approximately 37.5N. These springs were experimentally confirmed to withstand the weight of the tester.

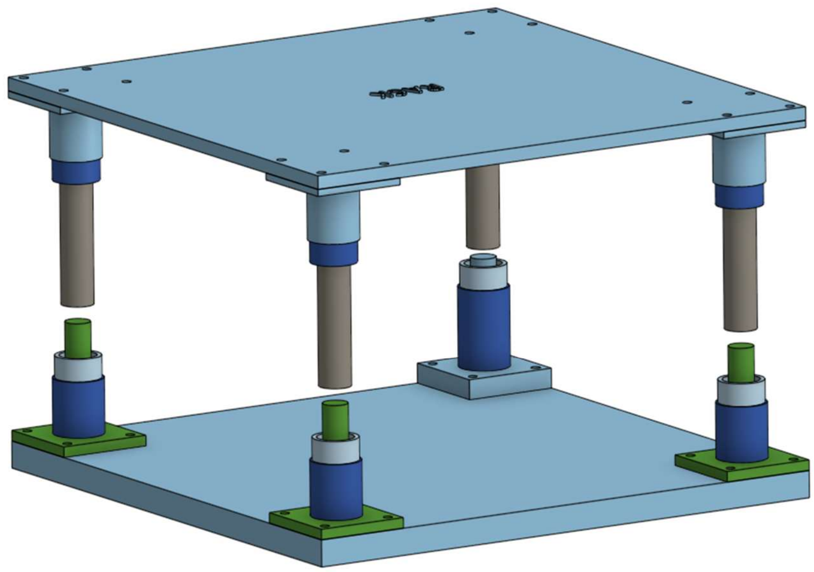

Figure 1: Digitally Base Assembly

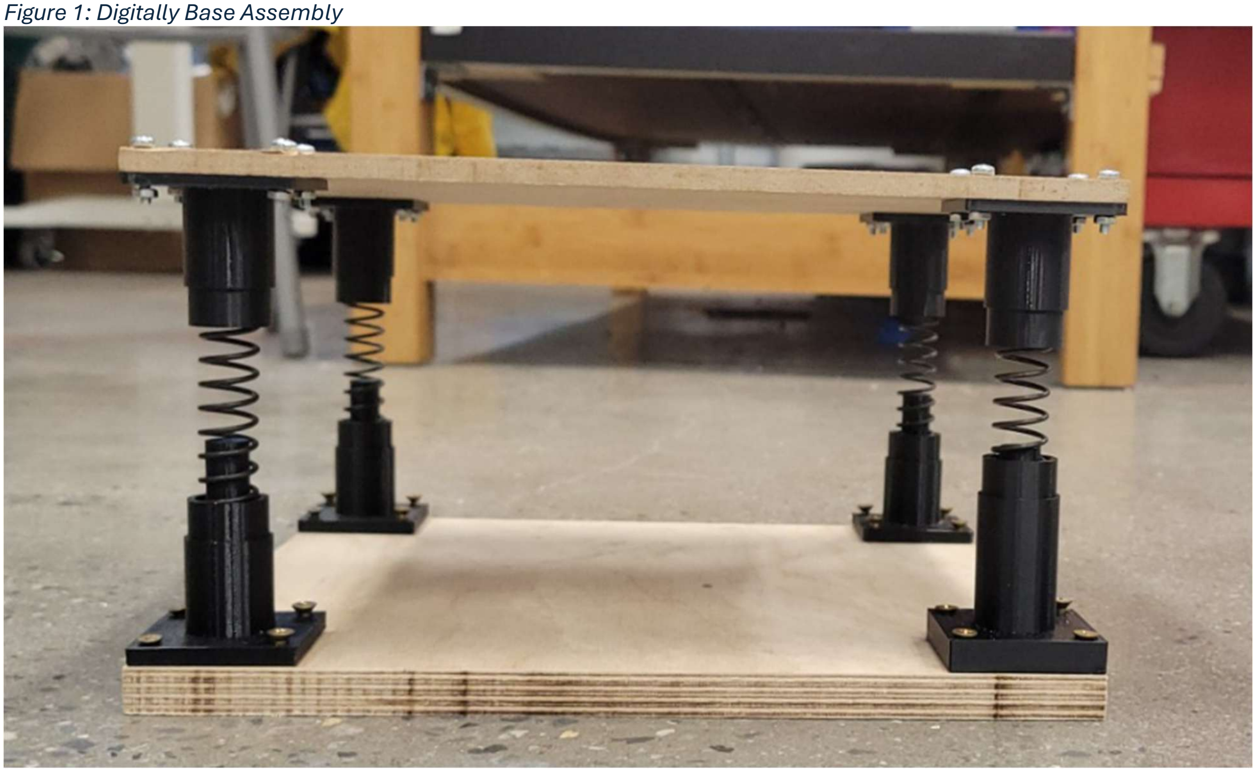

Figure 2: Base edition 1

The black components of figure 2 were created to mount the springs onto the tile. The black components were designed such that the springs sat inside of the components, and that the components themselves could be screwed into the board. This allowed for easy modifications upon future subsystems.

Revisiting: $F_s = -kx$ and rearranging for x(distance of spring compression) relates to the next subsystem.

Rack and Pinion:

The rack is a rectangle with teeth, that is capable of vertical motion. The teeth of the rack interlock with the Pinion. The Pinion is a gear with teeth designed to mesh with the rack. To enable rotation of the Pinion, mounts with holes for an axle were designed(figures 3 and 4 below).

Physical analysis/efficiency:

Calculating the x of the spring reveals the range of motion of the rack.

$$x = -\frac{F_s}{k} = \frac{-F_{applied}}{k} = -\frac{15N}{\frac{37.5N}{m}} = 0.4m = 4cm$$

The pinion was designed such that it was easy to laser print and scaled to the digital assembly, thus, the diameter of the pinion is 40mm=0.040m.

To calculate the revolution every time the tile is stepped on:

$$C = \pi d = 0.040\pi = 0.1256m$$

Since the rack moves 0.4m:

$$\frac{Rev}{step} = \frac{0.1256m}{0.4m} = 0.69\ revolutions$$

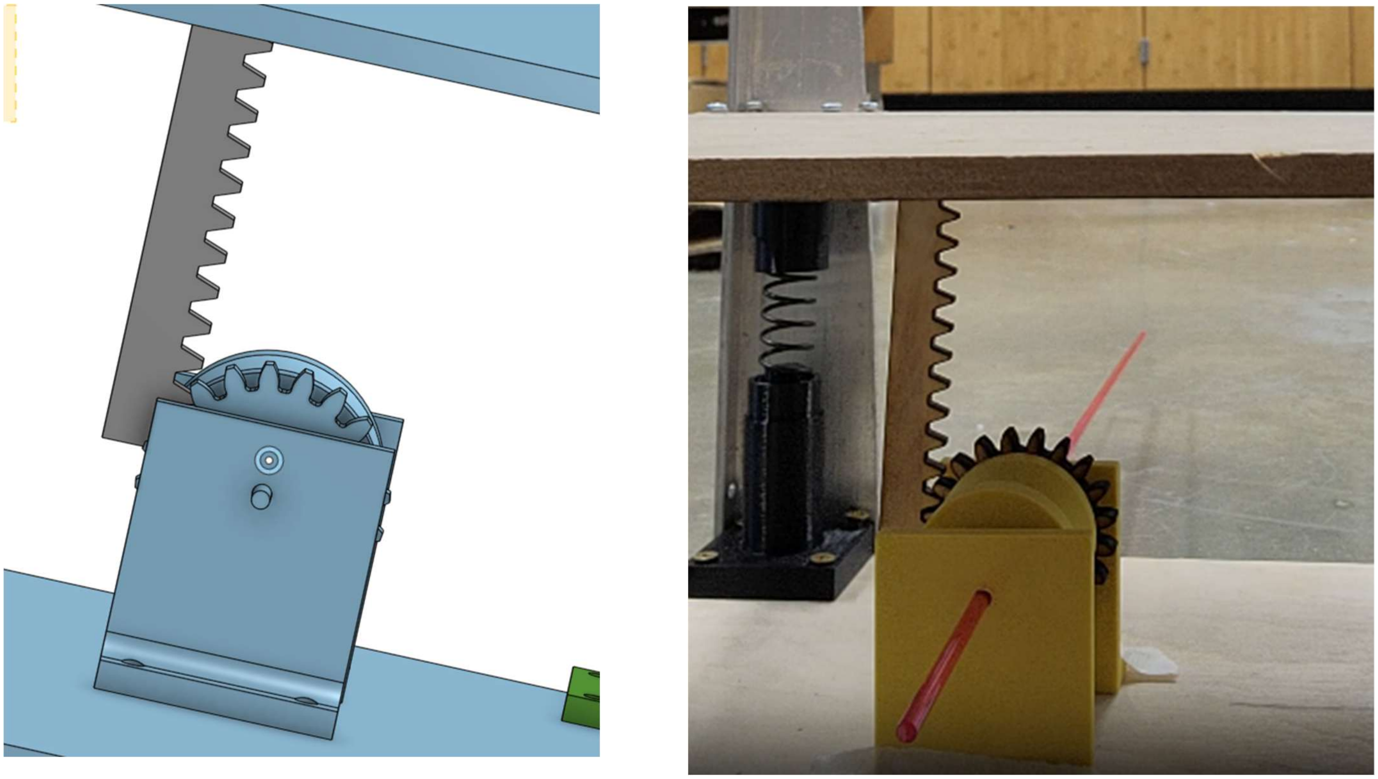

Figure 3: CAD

Note: The photo of the base above is edition 1, not the final edition, however, no photos were taken. Metal beams were added to prevent wobbling, which allowed for better meshing between the rack and pinion.

The 0.69 revolutions per step calculated above is impractical alone, however, that leads to the purpose of the next subsystem.

Pulley System:

Concentric holes were created in the pinion to allow for the pulley to mesh with the pinion. Using conservation of momentum and gear ratios, the 0.69 revolutions per step is increased through the pulleys. A second pair of pulley and gears was printed.

This time the gear has a diameter of 0.025m(the pulleys have the same diameters as the gears). Since the rack and pinion pulley has a diameter of 0.040m, the gear ratio is:

$$\frac{Pinion\ pulley\ diameter}{gear\ pulley\ diameter} = \frac{0.040m}{0.02} = 1.6$$

Therefore, the smaller pulley rotates 1.6 times faster than the pinion pulley. Thus, the number of revolutions must also increase. This makes the system more efficient.

The belt of the pulley system was designed such that the belt is an inextensible string. Since linear momentum could not be conserved if non-conservative forces such as the elastic force of a rubber band were acting on the system.

Linear Momentum:

$$P_{Before} = P_{After}$$

$$m_1 v_1 = m_2 v_2$$

Mass one = Mass two since this only involved the belt's linear motion.

Thus, the velocities must also be equal.

$$v_1 = v_2 = v$$

The angular momentum of the pinion pulley is conserved/transferred to the linear momentum of the "belt", ensuring constant linear velocity. This constant linear velocity enables the mechanical energy(kinetic energy) to be conserved and transferred to the smaller pulley.

The angular momentum of the pinion pulley is conserved; thus the angular velocity of the smaller pulley must be the same. Despite the velocity of the smaller pulley being the same, due to size ratio, the gear has more revolutions per minute.

Angular momentum expressions:

$$L_{pinion} = I_{pinion} w_{pinion}$$
$$I_{pinion} = \frac{1}{2} M_{pinion} R_{pinion}^2$$
$$L_{small} = I_{small} w_{small}$$

$$I_{small} = \frac{1}{2} M_{small} R_{small}^2$$

This leads to the Output subsystem.

Output:

The output consists of a gear(“output gear”) attached to the shaft of a 6V DC motor. The output gear meshes to the gear corresponding to the smaller pulley in the **pulley system**. Both gears have the same number of teeth and diameters, thus, the gear ratio is 1:1. As the gears spin, the shaft spins, which in-turn, produces the voltage.

To increase efficiency of this subsystem, an output gear with more teeth could have been printed, however, the 3D printer is not precise enough to print more teeth on such a small surface without flaw.